Mock Midterm Worked Solutions

Machine Learning in Engineering Science

Worked solutions and scoring guidance | Fall 2026 | Published 2 November 2026

Worked solutions and scoring guidance for the Mock exam. Attempt the 75-minute paper before reading. All numerical cases are synthetic teaching examples. The Mock itself has no course-grade weight.

Use the indicated points to self-assess a defensible method rather than memorise a preferred phrase. Equivalent terminology is acceptable when the conditional meaning is correct. Carry an isolated arithmetic error forward without repeatedly deducting for the same mistake. A correct number without requested reasoning does not earn reasoning marks.

The practice assesses target, split, baseline, evaluation and limitations. A reproducible but invalid evaluation remains invalid. No project demonstration, paid tool, code execution or guaranteed model-score improvement is required.

Question 1 Frame a vibration prediction experiment

20 points | Week 1 to 3 | Judgments: target, split, baseline, evaluation, limitations

Part a 4 points

1 point One pump run is a row; runs within a pump can be dependent.

1 point Target is vibration RMS ten seconds later in mm/s.

1 point The prediction is issued at run start with then-available inputs.

1 point This is supervised regression.

Part b 6 points

2 points Random row splits can share pump identities across partitions and fail to represent new-pump deployment.

2 points The global scaler uses held-out distribution information.

2 points Partition whole pump IDs into disjoint train, validation and test groups; fit preprocessing on train, select on validation, freeze and test. Accept a justified combined time/group protocol that preserves unseen-pump evaluation.

Part c 4 points

1 point Prediction = 0.5 + 0.05(30) + 0.2(2) = 2.4 mm/s.

1 point Signed residual = 2.4 - 3 = -0.6 mm/s.

1 point Absolute error = 0.6 mm/s.

1 point Squared error = 0.36 (mm/s) squared.

Part d 6 points

2 points The training target mean is the standard squared-loss constant baseline. Accept another explicitly training-only constant, including a median, with a defensible comparator claim; it need not be RMSE-optimal. Held-out targets must not determine it.

2 points Compare on identical held-out pump IDs using the same RMSE definition and permitted information.

1 point A stronger Ridge penalty shrinks coefficients and can trade variance for bias.

1 point It does not repair pump overlap or globally fitted preprocessing.

Marking cautions

A different random seed does not separate pump identities.

Question 2 Evaluate alarms under asymmetric costs

15 points | Week 5 | Judgments: evaluation, limitations

Part a 4 points

1 point TP = 12.

1 point FN = 8.

1 point FP = 6.

1 point TN = 54.

Part b 6 points

1.5 points Precision = 12/18 = 2/3 = 0.667.

1.5 points Recall = sensitivity = 12/20 = 0.600.

1.5 points Specificity = 54/60 = 0.900.

1.5 points F1 = 24/(24+6+8) = 12/19 = 0.632.

Part c 5 points

2 points At 0.4: 8(2)+20 = 36. At 0.7: 8(8)+6 = 70. One point per correct cost.

1 point Select 0.4 on validation.

2 points Freeze before an untouched test evaluation. Test predictions at 0.7 do not determine performance at 0.4. Do not tune thresholds using test outcomes; disclose reused tests and seek a new untouched final holdout if necessary.

Marking cautions

Keep validation and test evidence separate even when class counts coincide.

Question 3 Relate variance retention to prediction

10 points | Week 6 | Judgments: evaluation, limitations

Part a 2 points

2 points 1.6/(1.6+0.4) = 0.80 or 80 percent.

Part b 2 points

1 point PC1 score = 2/sqrt(2) = sqrt(2).

1 point PC1-only reconstruction = (1, 1).

Part c 2 points

2 points The discarded residual z - reconstruction = (1, -1) lies in PC2, the feature-difference direction. High retained input variance does not guarantee retention of the label signal.

Part d 4 points

2 points A factor-1000 change changes squared Euclidean distances and the raw-space clustering objective; use justified units/scaling.

2 points Cluster IDs are arbitrary names for geometric groups. Physical fault labels require independent validation.

Marking cautions

The reconstruction is not (2,0); the PC2 component has been dropped.

Question 4 Reason about a causal sequence model

25 points | Week 7 | Judgments: target, split, evaluation, limitations

Part a 4 points

1 point Input [1,3,2,5].

1 point Target x4 = 4.

1 point x4 is future and unavailable at time 3.

1 point Assign targets chronologically and construct all features from past/available information only; historical context can be used when legitimately known.

Part b 6 points

2 points Cell = 0.6(1)+0.5(0.8) = 1.0.

2 points Hidden = 0.25(0.7616) = 0.1904.

2 points Closing only the output gate gives hidden = 0, cell = 1 at this step. It does not erase memory, and later dynamics need not be unchanged.

Part c 8 points

3 points Masked exponentials [2,0,3], normalized weights [2/5,0,3/5].

3 points Context = (2/5)[5,0]+(3/5)[0,5] = [2,3].

2 points Unmasked softmax gives [2/6,1/6,3/6]; zeroing the middle entry leaves a sum of 5/6. Normalize over valid keys. Padding and causal masks address different exclusions.

Part d 7 points

2 points Output logits at positions 0 through 3 must agree within the agreed numerical tolerance.

1 point Each causal position can access only its own prefix; padding exclusion alone does not enforce that.

2 points Teacher forcing supplies known shifted inputs at all positions, permitting masked parallel operations; generation requires previously generated inputs.

2 points Position information gives ordering information absent from content-only attention; absolute or relative mechanisms can provide it.

Marking cautions

The middle key, not the last key, is padding here.

Question 5 Diagnose variation and a changed deployment

20 points | Week 9 | Judgments: evaluation, limitations

Part a 6 points

2 points Bias = 1.6 - 2.0 = -0.4; squared bias = 0.16.

2 points Affine MSE = 0.16 + 0.01 + 0.25 = 0.42.

2 points Nonlinear MSE = 0 + 0.08 + 0.25 = 0.33.

Part b 4 points

2 points Refit variance decreases from 0.25 to 0.01 while mean prediction/bias stay fixed.

1 point A curved mean residual can reflect a missing nonlinear relationship; more samples do not change the affine hypothesis class.

1 point Independent observation noise persists under the stated information assumptions.

Part c 4 points

2 points A is input/covariate shift with fixed P(Y given X).

2 points B changes P(Y given X); unchanged marginal inputs would not prevent this relationship shift.

Part d 6 points

2 points Test MAE = 0.2(3)+0.8(0.5) = 1.0.

2 points Reweighted MAE = 0.4(3)+0.6(0.5) = 1.5.

1 point The group-conditional mean absolute errors must remain stable.

1 point Changes within a group, target-response/noise changes or subgroup sampling uncertainty can invalidate the estimate. It is not direct deployment validation.

Marking cautions

MSE and RMSE are not interchangeable in the decomposition.

Question 6 Check predictions before claiming reproduction

10 points | Week 10 | Judgments: split, evaluation, limitations

Part a 3 points

1 point Check unique IDs and equal ID sets; reject duplicates/missing/extra IDs.

1 point Align by ID rather than row index.

1 point Aligned maximum absolute difference = 0 for each model.

Part b 2 points

2 points Equal rounded aggregate losses can conceal different individual predictions; compare every aligned value for both frozen models under a fixed tolerance.

Part c 2 points

2 points Reject before fitting. Relabelling rpm as revolutions per second changes the unit meaning without the required factor-60 conversion. A reviewed conversion would require explicit implementation outside this rpm-only contract.

Part d 3 points

1 point Matching hashes support byte integrity relative to the reference, and agreement supports this computational replay.

1 point They do not authenticate the sender or establish physical validity/generalization; require two explicit stronger claims, half a point each.

1 point Record a relevant Python/library/OS/runtime version; a new process alone does not prove an independently rebuilt environment or a different human.

Marking cautions

A data checksum is not a physical validation certificate.

Mock oral practice and scoring

Use a two-minute slot and the 12 + 10 + 8 point rubric below. These practice cases are provided for self-assessment and discussion.

Criterion	Points	Evidence
Decision and mechanism	12	6 for the justified central decision and 6 for its mechanism. Correct the claim rather than reciting unrelated definitions.
Evidence and next step or limit	10	4 for using the supplied evidence, 3 for a justified control or next step, and 3 for a concrete limitation.
Counterfactual reasoning	8	4 for an appropriate updated conclusion and 4 for the explanation. A yes or no alone is insufficient.

M1 Selection after test

Main question **Identify the main evaluation problem and propose a corrected next step.**

Fixed follow up **Would fixing the random seed make the final test claim valid?**

Decision and mechanism The test was used for model selection, so the reported minimum is not an untouched final estimate. Use training/validation for development, freeze the comparison and obtain new untouched final test evidence when justified.

Evidence and next step Five alpha choices were selected by the same test scores. The controlled procedure must preserve a baseline, declared metric and comparison population.

Counterfactual response A fixed seed improves repeatability of a computation, but does not undo information use or selection bias.

Claim boundary A reproducible optimistic result can still be an invalid generalization estimate.

Apply the common 12 + 10 + 8 point rubric. These checkpoints describe acceptable reasoning, not a required script.

M2 A unit label in a replay

Main question **Is this rejection useful or a model failure? Explain the next appropriate action.**

Fixed follow up **If you simply rename km/h back to m/s and obtain matching predictions, have you established that the original measurements were correctly encoded?**

Decision and mechanism This is useful contract enforcement before model fitting, not evidence that the fitted model failed. Diagnose the actual original units and make any justified conversion explicit and reviewed.

Evidence and next step Relabelled unchanged numbers change the interpretation. Keep the original artifact and record the failed attempt, correction and actual retest.

Counterfactual response No. Matching numbers after relabelling can reproduce an encoding error. Verify measurement provenance and units; do not silently alter metadata to force agreement.

Claim boundary Computational agreement does not establish the physical meaning of the original inputs.

Apply the common 12 + 10 + 8 point rubric. These checkpoints describe acceptable reasoning, not a required script.

Practice responses and partial credit

Case M1 strong response: The student selected alpha using the test scores, so the minimum is a development result, not an untouched final estimate. Keep a baseline, use training and validation to freeze the comparison, and obtain a genuinely untouched final holdout before a final claim. A fixed seed reproduces the same selection procedure; it does not undo its test information use.

Illustrative score for that response: decision and mechanism 12/12, evidence and next step or limitation 10/10, counterfactual 8/8, total 30/30. Equivalent concise reasoning can earn the same score; this is not a required wording.

Case M1 partial response: Use a fixed seed and run the code again. This recognises repeatability but does not diagnose test-driven selection or justify a new evaluation. Illustrative score: decision and mechanism 0/12, evidence and next step or limitation 2/10, counterfactual 0/8, total 2/30. Awarded points must follow the actual response, not the sample label.

Case M2 strong response: The loader is enforcing a unit contract before fitting. Preserve the rejected artifact, verify the source units, and record a justified conversion or correction followed by a real retest. Renaming the units and reproducing numbers does not validate the original physical encoding. This reasoning can earn all relevant points without any code execution.

Written practice scoring: sum the six raw question scores to 100. To rehearse the midterm combination, use 0.70 times that score plus one oral raw score out of 30. The Mock itself has no grade weight. A failed candidate improvement can still earn high marks when the design, evidence and limitations are sound.